Oncology Clinical Trials Diagnostic

Findings Briefing | November 2026 | Ilias Kubica

Situation

Trials that end early are costly for sponsors and for the people who take part. This diagnostic uses ClinicalTrials.gov data on 4,217 breast cancer trials completed or terminated since 2019 to ask where early termination is most common, and whether differences between countries and sponsors hold up once the kind of trial is taken into account. Overall, 15.1% of trials were terminated (635 of 4,217).

Finding 1: Trial type drives termination, so raw comparisons mislead

Termination rates are 28.6% for Phase 1 and 26.7% for Phase 2 trials, against 7.7% for non-drug trials. Industry sponsors terminate 25.6% of their trials, against 11.9% for academic, hospital and other sponsors. Most countries (24 of 32 with 50+ trials) sit between 16% and 27%. Australia is highest at 26.8% (149 trials), but with a margin of error of about 7 points it is not clearly different from the US (22.6%, 2,023 trials).

Finding 2: Turkey and Egypt are low, and trial mix explains only part of it

Turkey terminated 7 of 244 trials (2.9%) and Egypt 0 of 135. At the overall rate, Turkey would have about 37 terminations and Egypt about 20. Both countries run more non-drug trials than average, so applying the overall termination rate for each phase to their phase mix gives an expected 25 for Turkey and 17 for Egypt. Phase mix therefore accounts for roughly 40% of Turkey's shortfall and about 15% of Egypt's, and most of the gap remains. Neither country has a single terminated academic or hospital trial (0 of 303 combined, where about 36 would be expected at the overall rate), and all seven of Turkey's terminations are industry-sponsored. This analysis cannot say why. Differences in how trials are registered and updated are one possibility worth checking.

Finding 3: Sponsors with similar volume differ widely

Among sponsors with 20 or more trials, termination rates run from 0% (Fudan University, Ohio State, City of Hope) to 39% (Washington University, 28 trials). Memorial Sloan Kettering (2 of 67, 3.0%) and M.D. Anderson (21 of 70, 30.0%) have almost identical volumes and the same sponsor type. Among large industry sponsors, AstraZeneca terminates 3.3% of trials against 21.3% for Novartis. I have not tested whether phase mix explains these gaps.

Finding 4: Countries with longer trials terminate more

Across the 32 countries with 50 or more trials, average trial duration and termination rate have a correlation of 0.55. Turkey and Egypt are the two shortest and lowest points and pull on that figure; I have not tested it without them.

Recommendation

Compare termination rates only after accounting for trial phase and sponsor type, because raw country and sponsor league tables mostly reflect what kind of trials each runs. Where a country still looks like an outlier after adjustment, as Turkey and Egypt do, check how trial status is recorded before reading it as performance. Test the sponsor gaps, such as M.D. Anderson against Memorial Sloan Kettering, for phase mix before drawing conclusions.

Method and limits: *ClinicalTrials.gov snapshot taken in October 2026, covering trials completed or terminated since 2019, by lead sponsor only. Multinational trials count in every country where they have a site, so countries are not independent. Non-drug and phase-not-reported trials are kept as separate groups. A counting error (site rows counted as trials) was found and fixed during the project, and earlier figures were corrected. Findings are associations, not causes.*

Code, SQL and dashboard: github.com/iliaskubica/clinical-trials-diagnostic